Forecasting district wheat yield

System architecture and component specification

A probabilistic forecast for 119 Indian districts, issued on 5 March —
eight weeks before the harvest it predicts.

THE TASK

Predict the harvest before it happens

One number is not enough. The system outputs a full range of possible harvests and how likely each is.

Predict	wheat yield in kg/ha	for each district, each season
When	5 March	eight weeks before harvest
Where	119 districts	Haryana, Punjab, Uttar Pradesh
Typical yield	≈ 4,500 kg/ha	median district area 3,405 km^2

The hard constraint

Nothing observed after 5 March may enter the forecast — no weather, no satellite image. Weather forecasts must have been published at least two days earlier. This is enforced in code, not by convention.

THE TASK

What the system produces

For every district and every season, a single row containing all of this.

Point forecast	4,162 kg/ha	the single best estimate
19 quantiles	q05 … q95	the full range of plausible harvests
P(yield rises)	8%	chance of beating last season
P(fall > 5%)	75%	chance of a mild decline
P(fall > 10%)	46%	chance of a serious decline
P(fall > 20%)	4%	chance of a severe decline

Values shown are the actual forecast for Rewari district, Haryana, in 2022 — used as the worked example later. Its harvest came in at 4,150 kg/ha.

ARCHITECTURE

The complete system

System architecture — the 5 March forecast

INPUT DATA

Yield history 1990–2022
MODIS satellite 2000–2022
Sentinel-2 2017–2022
Daily weather 2010–2023
10-day forecast, MSP price

STAGE 1 — Blend

H weighted history 0.50
R weather+satellite 0.15
P physics 0.00
E economic 0.15
C global transfer 0.20

+ disagreement gate
+ movement scale ×1.50

STAGE 2 — Weather ahead

four matched models, differing only in whether the forecast is visible

NO-FUTURE = control
FULL / EFFECT / BROAD

c = mean(3) − control ×1.75

STAGE 3 — Training data

same model, two spans of training seasons

from 2013 → 952 rows
from 2017 → 238 rows

c = wide − narrow ×0.25

STAGE 4 — Crop vision

transformer turns satellite crop state into 16 values

two matched models: with them, and without

c = with − without ×2.25

FEATURE PANEL

78 numeric inputs
yield history 18
observed weather 16
forecast weather 10
stress indices 31
economics 3
+ 3 state indicators

POINT FORECAST

$\hat{y}$ = Blend + 1.75·c(ahead)
+ 0.25·c(data) + 2.25·c(crop)

each c is a matched-model difference —
two identical models, one extra input

STAGE 5 — Uncertainty

empirical distribution of the model's own past errors, scaled by district volatility, seasons weighted equally, recentred on $\hat{y}$
→ 19 quantiles q05 … q95

BASELINE b

b = 0.60·y(t−1)
+ 0.25·y(t−2)
+ 0.15·y(t−3)

models predict r = y − b

OUTPUT — per district, per season

point forecast (kg/ha) · 19 quantiles q05 … q95 · P(yield rises)
P(fall > 5%) · P(fall > 10%) · P(fall > 20%) · P(fall > 30%)

every probability read off the interpolated cumulative distribution

ARCHITECTURE

How the pieces fit together

The system is a conservative estimate, adjusted four times. Nothing replaces the estimate; each stage only nudges it.

```
ŷ  =  Blend   +   1.75 · c(weather ahead)   +   0.25 · c(training data)   +   2.25 · c(crop vision)
```

Stage 1	Blend	five simple estimates averaged, with a gate and a movement scale
Stage 2	Weather ahead	what changes when the model may see the 10-day forecast
Stage 3	Training data	what changes when the model is trained on four more seasons
Stage 4	Crop vision	what changes when a network summarises satellite crop condition
Stage 5	Uncertainty	turns the single number into 19 quantiles and every probability

Each “c” is a correction. The next slides define each one precisely.

INPUTS

Where the information comes from

Source	Coverage	Volume	What it contributes
Harvest records	1990–2022	3,658 seasons	ground truth and history
MODIS satellite	2000–2022	2,737 × 3 × 35	crop condition, coarse
Sentinel-2 satellite	2017–2022	2,142 × 126	crop condition, fine
Daily weather	2010–2023	546,805 days	temperature, rain, sunlight
Weather forecast	per season	10 × 5-day windows	conditions after the clock
Support price	annual	3 features	economic signal

Satellite indices measure how the crop reflects light: healthy plants reflect infrared strongly and red weakly, so the ratio tracks living plant material.

INPUTS

The 78 numbers the models actually see

Raw data is compressed into a fixed panel of 78 columns, computed identically for every district and season.

18 Yield history	lags 1–5, five- and ten-year summaries, state mean, district minus state
16 Observed weather and soil	early, mid and late season windows; soil moisture at three depths
10 Forecast weather	temperature, rainfall and solar anomalies, plus bias-corrected variants
31 Stress indices	heat degree-days above 32 °C and 34 °C, water balance, district and regional
3 Economics	support price level, one-year and three-year changes

Plus three binary state indicators, and a flag marking any value that was missing and had to be filled in.

STAGE 0

The starting estimate

Before any model runs, take a weighted average of the district's last three harvests.

$$b = 0.60 \cdot y(t-1) + 0.25 \cdot y(t-2) + 0.15 \cdot y(t-3)$$

Why this works

- Yield is persistent — irrigation, soil and farming practice change slowly
- A district that produced 4,500 kg/ha for three years will probably do so again
- Weights decay because recent seasons are more informative

What every model predicts instead

$$r = y - b$$

No model ever predicts yield directly. Each predicts the residual — how far this season departs from business as usual.

Why that matters more than it looks

No model has to learn that one district out-yields another — the baseline absorbs it. Every bit of learning capacity is spent on what is unusual about this season, which is the part that actually varies.

STAGE 1

The Blend — five estimates, averaged

Five separate models look at the season differently. Averaging cancels their independent mistakes.

H Weighted history	the baseline from the previous slide	0.50
R Weather + satellite	trees on observed weather and satellite to 5 March	0.15
E Economic	trees on lagged prices and support-price changes	0.15
C Global transfer	a model trained on wheat elsewhere in the world	0.20
P Physics	soil water balance and heat stress	0.00

The physics member has weight zero — it did not improve the average. It is kept because its direction still votes in the gate on the next slide, and a direction can be informative even when a level is not.

STAGE 1

The disagreement gate

Normally trust history. But when the other models agree that this season is different, stop leaning on last year.

```
s = sign( member − last season )        for R, P, E and C

S = s(R) + s(P) + s(E) + s(C)
```

The gate fires when both are true:

- $|S| \geq 2$ — a majority agree on direction
- that direction contradicts what history implies

Weights switch

```
normal   0.50 H · 0.15 R · 0.15 E · 0.20 C
fires    0.20 H · 0.30 R · 0.30 E · 0.20 C
```

In practice

The gate fires on 20.6% of district-seasons. On the other 79.4% the blend stays anchored to history, which is the safe default when nothing unusual is being signalled.

STAGE 1

Movement calibration

Averaging five models pulls the result toward the middle. The blend under-moves, so its movement is scaled back up.

$$\text{Blend} = L + 1.50 \cdot (\text{Blend\_raw} - L) \qquad \text{where } L = \text{last season}$$

How the 1.50 was obtained

Take historical cases. Let x be how far the raw blend moved from last season, and z how far the truth actually moved. Fit the single number that best rescales one into the other, by least squares through the origin:

$$s = \Sigma\, x_i z_i \;/\; \Sigma\, x_i^2$$

What it means

The answer came out near 1.5.

When the blend says “up 100 kg/ha”, the truth has historically moved about 150. The scale undoes the shrinkage that averaging introduced.

Stage 1 produces one conservative yield estimate per district-season. Stages 2 to 4 adjust it; none of them replaces it.

STAGES 2–4

How each correction is measured

Train two models that are identical in every respect — same rows, same target, same settings, same random seeds — except that one is given an extra piece of information.

The difference between their predictions is what that information contributes, and nothing else.

This is the same logic as a drug trial: two groups treated identically apart from one thing, so any difference is attributable to that thing.

Stage 2	the 10-day weather forecast	`mean(FULL, EFFECT, BROAD) − NO-FUTURE`	×1.75
Stage 3	four more seasons of training data	`trained-from-2013 − trained-from-2017`	×0.25
Stage 4	16 values from the crop network	`with-those-values − without`	×2.25

Each weight is deliberately less than one: a correction that cannot be measured precisely is applied at a fraction of its face value, so noise cannot dominate.

STAGES 2–4

The prediction models underneath

Every correction is produced by gradient-boosted decision trees — hundreds of small yes/no question trees, averaged.

Settings

Trees	`350`
Maximum depth	`2`
Learning rate	`0.025`
Minimum child weight	`25`
Row subsample	`0.85`
Column subsample	`0.65`
L2 penalty	`50`
L1 penalty	`5`
Random seeds	`42 and 73, averaged`

Why depth 2

Each tree may ask only two yes/no questions. With few independent harvests to learn from, a deeper tree would memorise districts instead of learning how they respond.

Preparation

- Missing values filled with the column median, plus a flag
- State encoded as three binary columns
- Predictions clipped to 500–7,000 kg/ha

Identical settings across every matched pair, so a measured difference can only come from the extra information — never from a different configuration.

STAGE 4

Why a neural network at all

The satellite sees the crop. The question is how to turn a large, messy pile of measurements into a few useful numbers.

The obvious approach

Compute summary statistics — average greenness, its spread, its 10th percentile. This works, but the recipe is fixed by the analyst in advance, and it treats every part of a district as interchangeable.

What a network does instead

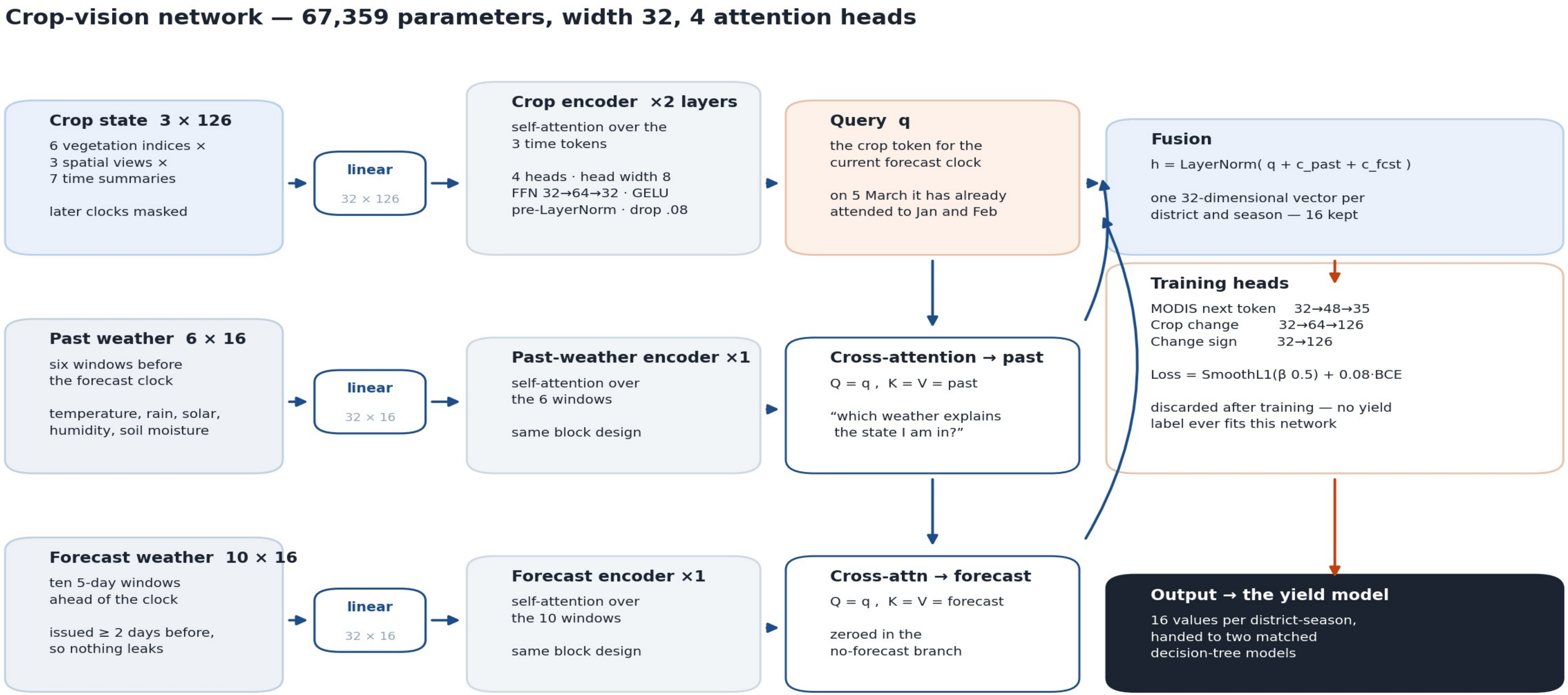
It learns the recipe. Formally it is a function that maps the whole pile to 16 numbers — and the entire design question is which 16.

The decision that shapes everything: it is never shown a harvest figure

There are thousands of examples of how a crop changes between January and March, and only a handful of harvests. So the network is trained on the data-rich question — how does wheat develop — and its internal summary is handed to a simple model for the data-poor question of what the yield will be.

STAGE 4

The crop-vision network



STAGE 4

What the network is shown

Three streams of numbers. Each row of a stream is called a token — the unit the network reasons over.

Crop state	**3 × 126**	6 vegetation indices × 3 spatial views × 7 time summaries, at 15 January, 15 February and 5 March
Past weather	**6 × 16**	six windows before the clock: temperature, rainfall, solar radiation, humidity, wind, soil moisture at three depths
Forecast weather	**10 × 16**	ten dated five-day windows ahead, with spatial spread and counts of days above 32 °C and 34 °C

Masking — how the leakage rule is enforced inside the network

At the 15 February clock the March token does not yet exist. Its attention score is set to minus infinity, so it receives exactly zero weight and cannot influence anything.

STAGE 4

Attention — the one operation to understand

Each token looks at all the others, decides how much each one matters, and takes a weighted average.

$$\text{Attention}(Q, K, V) = \text{softmax}(Q \cdot K^T / \sqrt{8} + M) \cdot V$$

$Q \cdot K^T$	every token's question is compared against every token's label — a large value means “you are relevant to me”
$\div \sqrt{8}$	keeps those numbers in a sensible range as the network gets wider
$+ M$	the mask: minus infinity for anything not yet observable, so it gets zero weight
softmax	turns the scores into weights that are positive and add up to one
$\cdot V$	the output is a weighted average of the other tokens' content

Four of these run in parallel with separate learned weights, so the network can attend to several different things at once.

STAGE 4

The crop asks the weather two questions

The crop token for the current date queries each weather stream separately.

```
c_past = MHA( q , past )
```

"Which past weather explains the state I am in?"

```
c_fcst = MHA( q , forecast )
```

"Which forecast weather threatens what happens to me next?"

```
h  =  LayerNorm(  q  +  c_past  +  c_fcst  )
```

LayerNorm rescales the vector to a stable range. Adding q back is a residual connection: information can bypass a layer that does not help it.

Result

one 32-value vector per district-season; 16 are kept

STAGE 4

How the network is trained

Two steps, neither of which uses a harvest figure.

Step 1 — learn general crop development

On coarse satellite data back to 2000, predict the next observation from the previous ones.

`AdamW · lr 8e−4 · 24 epochs`

Step 2 — specialise on recent, sharper data

On fine satellite data from 2017, predict how each of the 126 crop measurements will change.

`AdamW · lr 6e−4 · 45 epochs · seeds 42, 73`

```
Loss = SmoothL1( predicted change, actual change ) + 0.08 · BCE( predicted direction )
```

Smooth L1 behaves like squared error for small mistakes and like absolute error for large ones, so a cloud or a sensor fault cannot dominate training. Targets are standardised using the median and median-absolute-deviation for the same reason.

PROTOCOL

Preventing the model from cheating

Two mechanisms, both essential when the same districts appear in training and testing.

Time cutoffs

- For a target season, nothing from that season or later is used anywhere
- This includes the averages used to standardise inputs
- Weather forecasts must predate the clock by two days

District cross-fitting

Districts are split into three groups. The 16 network values for a district in group 1 are produced by a network trained without group 1.

Otherwise those values would look informative simply because the network had memorised the district.

Evaluation protocol

- Rolling origin — the model for each season is trained only on earlier seasons
- Correction weights are chosen on development seasons and then held fixed
- Later seasons are used once, for confirmation, never to choose anything

STAGE 5

Turning one number into a distribution

Rather than assuming a bell curve, use the errors the model has actually made in the past.

1	Collect past errors	`e = y − ŷ, using only seasons before the target`
2	Put them on a common scale	`a = max( recent SD, 7% of normal yield, 150 )    u = e / a`
3	Weight seasons equally	`each row gets weight 1/n(t), so a big season cannot dominate`
4	Take weighted quantiles	`interpolate u at the 19 probability levels`
5	Rescale and recentre	`Q(α) = ŷ + a · u(α)`
6	Enforce validity	`running maximum so quantiles cannot cross, then clip`

Step 2 matters: a 300 kg/ha error means something different in a stable district than a volatile one, so errors are divided by a district-specific size before being pooled.

STAGE 5

From quantiles to answers

The 19 quantiles are the distribution. Every question anyone asks is a lookup on it.

The quantiles define F

```
F( Q(α) )  =  α
```

F is the chance the harvest comes in at or below a given yield. It is known at 19 points and filled in between by straight lines.

Every question is a lookup

```
P( rises )      = 1 − F( last )
P( fall > p )   = F( (1−p) · last )
```

Published at 5%, 10%, 20% and 30%.

The density curve is the slope of F

Where quantiles bunch together, a lot of probability sits in a narrow band of yields, so the curve is tall there. Where they spread out, the curve is low.

Output: one row per district-season with the point forecast, all 19 quantiles, and every threshold probability.

OPERATION

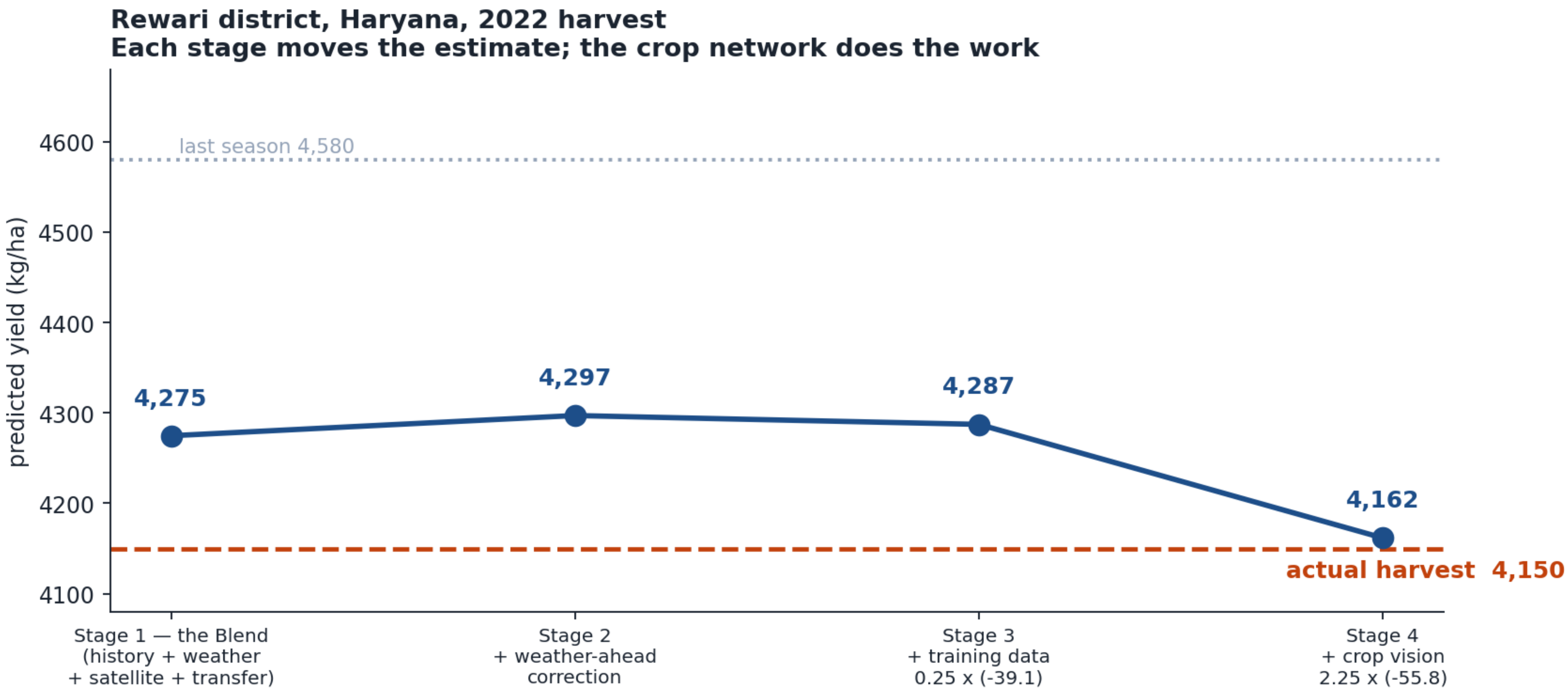
Running the model for one district

Nine steps, roughly three hours end to end from raw data on a single laptop.

1. Assemble lagged yield history and compute the baseline
2. Compute the 78 features from weather, soil, satellite and price data
3. Evaluate the five Blend members, apply the gate, apply the ×1.50 movement scale
4. Run the four weather-forecast models; take their difference; add at ×1.75
5. Run the two training-window models; take their difference; add at ×0.25
6. Build the crop tensors, mask anything after the clock, run the network, take 16 values
7. Run the two matched tree models with and without them; add the difference at ×2.25
8. Scale the historical error distribution to the district and recentre it
9. Interpolate the quantile curve and read off every threshold probability

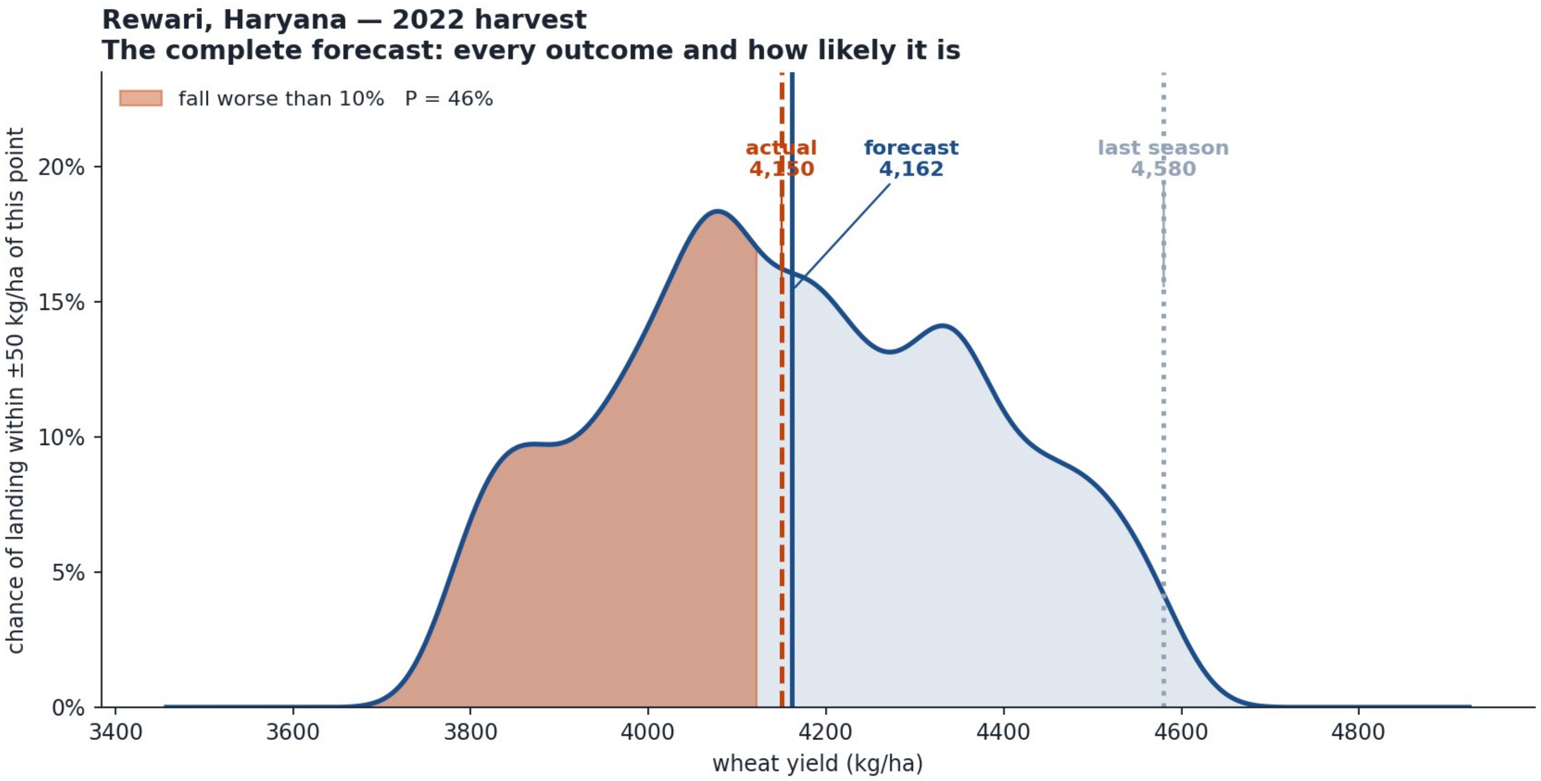
WORKED EXAMPLE

Rewari district, Haryana — 2022



WORKED EXAMPLE

The forecast that is issued



Point forecast	**4,162**
80% range	**3,822 – 4,529**
P(increase)	**8%**
P(fall > 5%)	**75%**
P(fall > 10%)	**46%**
P(fall > 20%)	**4%**

Actual harvest 4,150

A 9.4% fall. The forecast placed it at the 10% boundary and put a decline at 92%.

RESULTS

Accuracy of the 5 March forecast

Measured on seasons the model never saw during fitting.

Model	2019–20	2021–22	four-year	direction
Baseline (3-season average)	–	–	**333.1**	–
Stage 1 — Blend	257.0	288.6	**273.3**	77.9%
+ Stage 2 weather ahead	254.1	288.1	**271.7**	78.8%
+ Stages 3 and 4 — full model	248.6	281.2	**265.4**	78.8%

Typical error of 265 kg/ha is about 5.9% of a 4,500 kg/ha harvest. Direction accuracy is the share of districts where the forecast landed on the correct side of last season.

RESULTS

The same system run seven weeks earlier

A forecast can also be issued on 15 January. It is usable, and clearly weaker.

Forecast date	2019–20	2021–22	four-year	direction
15 January	296.8	316.9	307.0	70.2%
5 March	248.6	281.2	265.4	78.8%

What waiting buys

- 41.6 kg/ha less error — about 13% better
- 8.6 more percentage points of direction accuracy
- The crop has entered grain filling and is visible to the satellite

Both dates are honestly calibrated

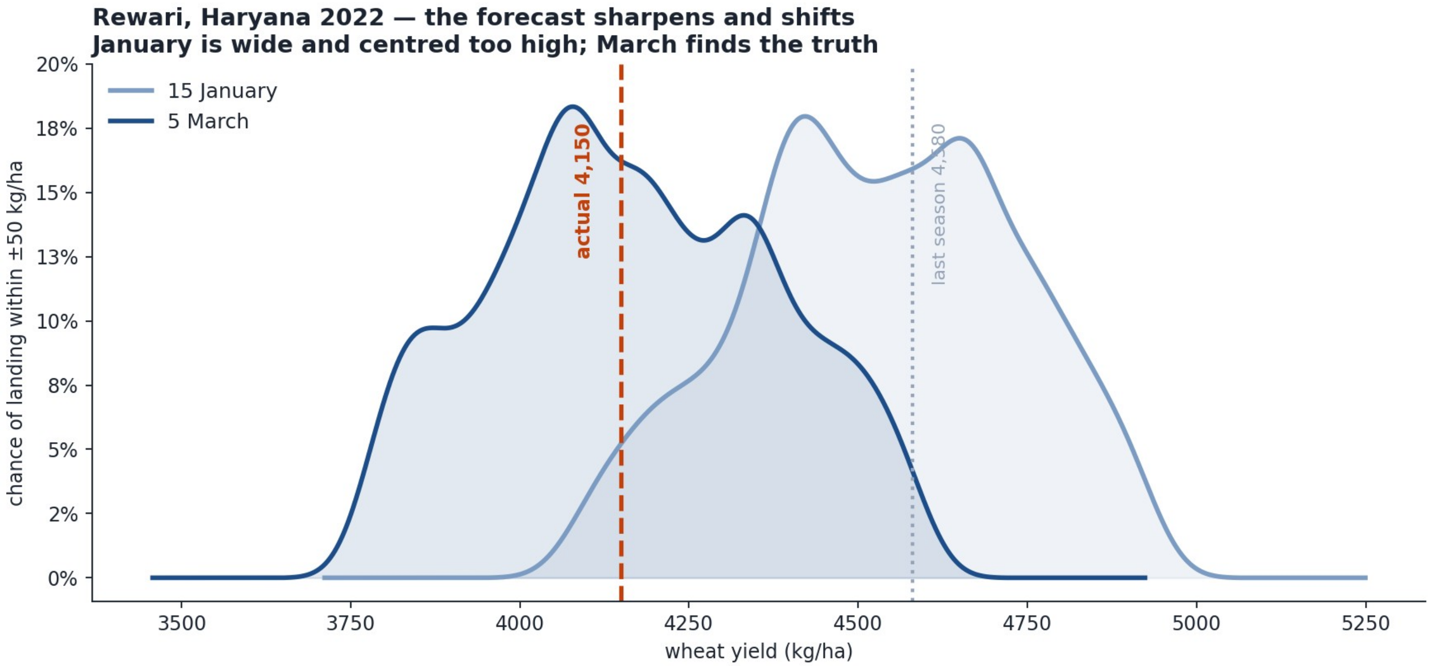
	15 Jan	5 Mar
80% ranges cover	79.3%	80.9%
90% ranges cover	90.5%	91.8%

Both should read 80% and 90%.

RESULTS

One district, both forecast dates

Rewari 2022. Last season 4,580 kg/ha; the harvest came in at 4,150.



	15 Jan	5 Mar
Forecast	4,555	4,162
Error	405	12
P(increase)	43%	8%
P(fall > 10%)	9%	46%

In January the model expected a good year and put a serious decline at 9%. By March it had seen the crop, moved the forecast down 393 kg/ha, and raised that probability to 46%.

RESULTS

Are the probabilities honest?

A probability is only useful if it is true. Across all 476 district-seasons of the 5 March forecast:

When the model says…	stated	actually happened
Fall worse than 5%	29.9%	29.2%
Fall worse than 10%	11.5%	10.9%
Fall worse than 20%	1.2%	2.3%
Any increase	45.9%	43.5%

80.9%

of harvests land inside the 80% range

An event the model calls 10% likely happens on about 10% of occasions. That is what makes these numbers usable for a decision rather than decoration.

One honest exception: the extreme tail is mildly over-confident — 1.2% stated for declines worse than 20%, against 2.3% observed.

The architecture in four sentences

A conservative baseline

a weighted average of the last three harvests; every model predicts the residual around it, so none has to learn what a district is normally worth

Four corrections

each measured as the difference between two otherwise identical models, and each deliberately shrunk before it is applied

One small network

67,359 parameters, trained on how crops develop rather than on yield, contributing 16 numbers to the final estimate

An empirical distribution

the model's own past errors, rescaled and recentred, giving 19 quantiles and every threshold probability

Output: a point forecast, a calibrated distribution, and an early warning of collapse — for 119 districts, eight weeks before harvest.